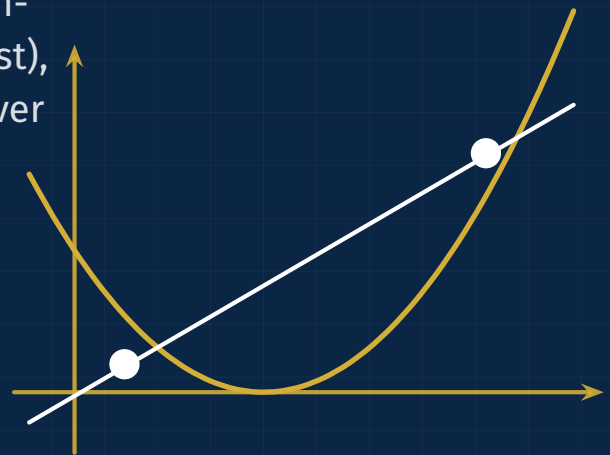


DR ESLAM AHMED
eliteigcse.com

CHAPTER 3 • SEQUENCES, FUNCTIONS & GRAPHS • TOPIC 3

Functions

A function is a rule: put x in, get one answer out. Substitute, combine (inner first), reverse (swap and rearrange) — and never divide by zero.

**6**

VARIANTS

39

PAST-PAPER PARTS

47

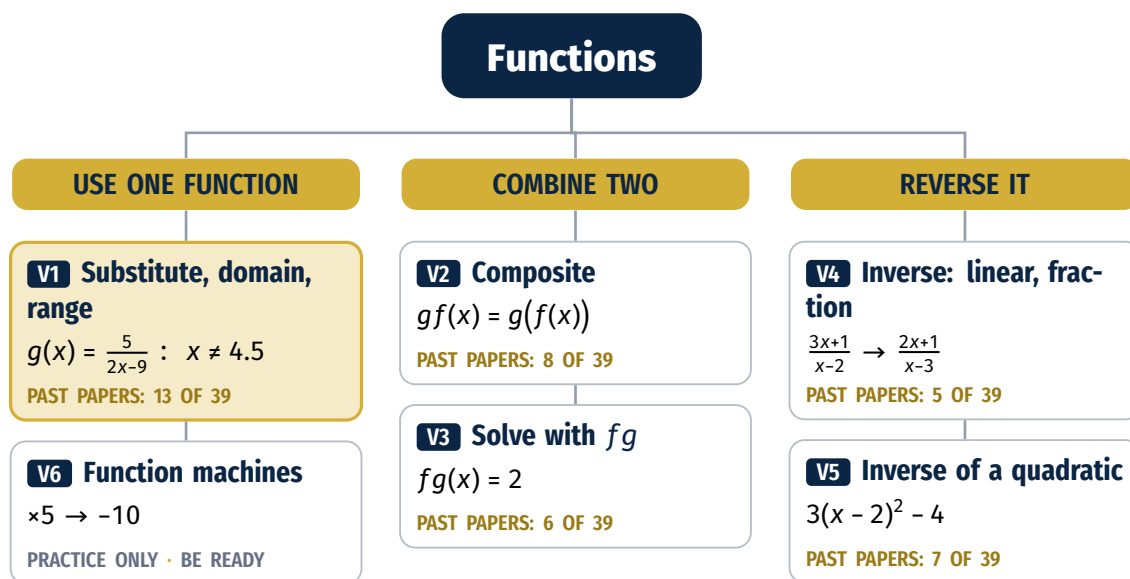
PRACTICE QUESTIONS

YOU WILL MASTER**V1** Substitute, domain & range**V2** Composite**V3** Solve with fg **V4** Inverse: linear & fraction**V5** Inverse of a quadratic**V6** Function machines

TOPIC MAP • EVERY VARIANT ON ONE PAGE

Functions at a glance

 Your notes



★ How often does each variant come up?

19 past-paper questions (Jan 2020 → Nov 2025) hold 39 parts; 47 practice parts — grouped by type.

	PAST PAPERS (OUT OF 39)	PRACTICE PARTS
V1 Substitute, domain, range	13	10
V2 Composite	8	14
V3 Solve with fg	6	9
V4 Inverse: linear, fraction	5	5
V5 Inverse of a quadratic	7	2
V6 Function machines	0	7

WHAT THIS TELLS YOU Functions are on every paper as one long question (Q15–25, 3–9 marks) made of small parts: an excluded value, an $fg(2)$, a $gf(x)$, an inverse. Each part is a separate, quick mark.

BEFORE YOU START

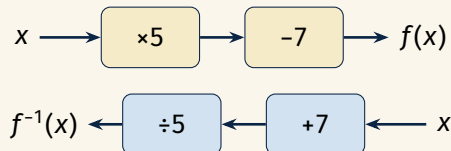
The big idea



A function is a machine.

$f(x) = 5x - 7$: whatever goes in the bracket replaces **every** x . $f(2) = 3$, $f(a + 1) = 5a - 2$.

$fg(x)$ means do g first, then f . f^{-1} runs the machine backwards.



The vocabulary

Domain

$$x \geq 1$$

The x values allowed in.

Range

$$f(x) \geq 0$$

The values that come out.

Excluded

$$\frac{11}{2x-5} : x \neq 2.5$$

Bottom = 0 is not allowed.

Composite

$$fg(x) = f(g(x))$$

Right-hand letter first.

 Read fg from the inside


CALCULATOR FIRST, THEN CHECK

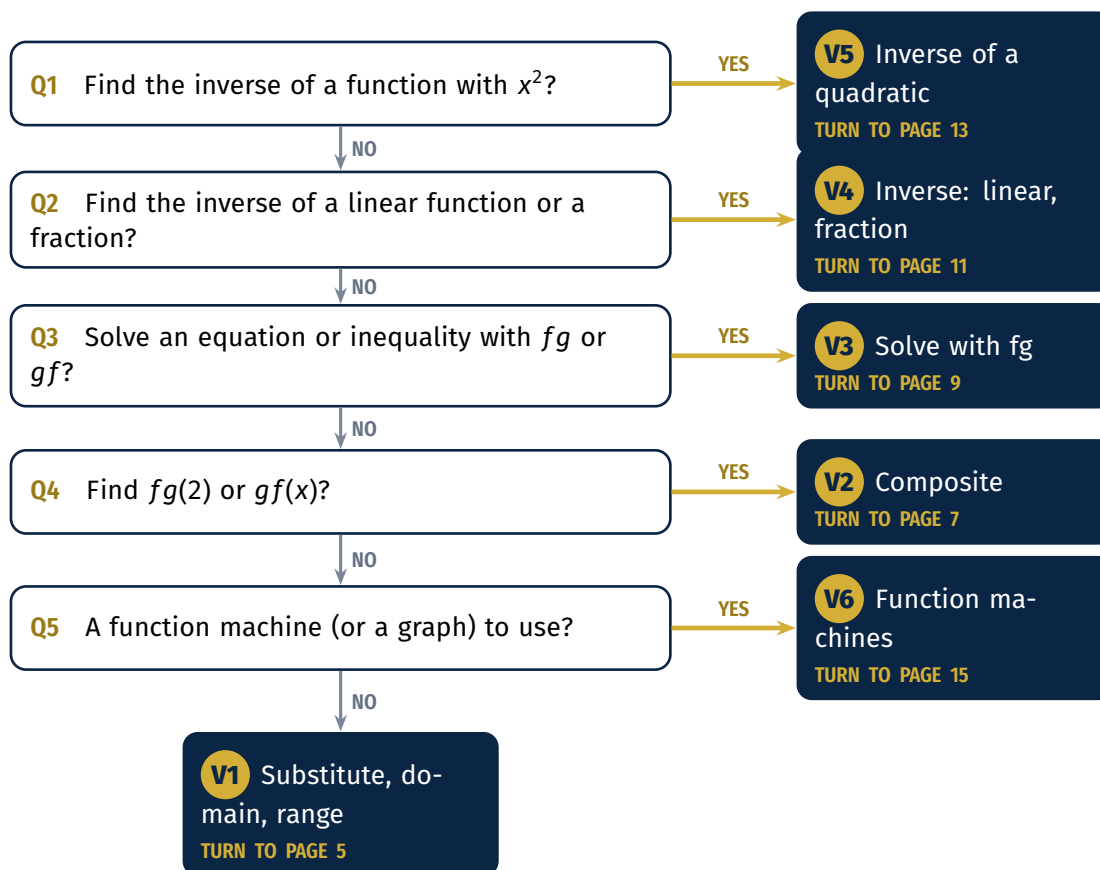
Check an inverse: $f(f^{-1}(x)) = x$. Try a number: $f(2) = 3$, so $f^{-1}(3)$ must be 2.

COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?

 Your notes

Read the question, then walk down the ladder. The first **YES** tells you the variant — and the page to turn to.



COLOUR CODE IN THE WORKING

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of “How do I do it?” each line is.

ON EVERY VARIANT SPREAD



What is it?



How do I do it?



Worked example



Check yourself



When you see → you do



Watch out



Exam tip



Calculator check

V1

VARIANT 1 MOST COMMON IN THE EXAM

Substitute, Domain & Range

Replace every x ; the bottom can't be 0; a square can't be negative.

SEEN IN PAST PAPERS

13 / 39

PRACTICE QUESTIONS: 10

 Your notes

Came up in: Jan 2020 P1HR Q17 • Oct/Nov 2020 P1HR Q16(a) • Jan 2021 P2HR Q17(c) • Nov 2021 P2H Q15(a) • Jan 2022 P2H Q25(b) • Jun 2022 P1H Q14(a) • May 2023 P1H Q17(a) • Nov 2023 P2H Q19(a) • May 2024 P1H Q15(a) • Jun 2025 P1H Q16(a) • Jun 2025 P2HR Q19(a) • Nov 2025 P2H Q19(a)

? What is it?

In plain words

Substitute: put the number in a bracket wherever x is.

Excluded value: a fraction can't have 0 on the bottom — solve bottom = 0. A square root can't take a negative.

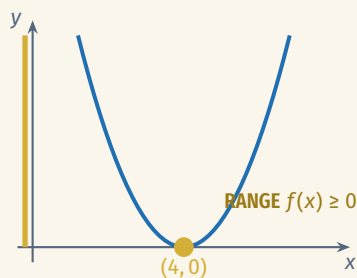
Range: what can come out? $(x - 4)^2$ is never below 0.

IT LOOKS LIKE

$$f(1)$$

$$\frac{5}{2x-9}$$

$$\text{range of } (x - 4)^2$$



How do I do it?

1 Substitute

Bracket the number: $f(1) = (1 - 4)^2$.

2 Excluded

Bottom = 0: solve it.

3 Range

Smallest (or largest) output: squares ≥ 0 .

4 Domain

The x values given (e.g. $x \geq 3$).

BOTTOM $\neq 0$ The quickest mark on the paper.

V1 Substitute, Domain & Range

VARIANT 1 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

"State the value of x that cannot be included in any domain of g " → **Bottom = 0, solve**

"State the range of the function f " → **Smallest possible output**

"Find $f(10)$ " → **Bracket and substitute**

WORKED EXAMPLE **PAST PAPER**

January 2020 · Paper 1HR · Q17(a)(b) · 2 marks

The function f is such that $f(x) = (x - 4)^2$ for all values of x . (a) Find $f(1)$. (b) State the range of the function f .

STEP 1 $f(1) = (1 - 4)^2$ ◀ bracket the 1

STEP 2 $= (-3)^2 = 9$ ◀ (a)

STEP 3 $(x - 4)^2 \geq 0$, equal to 0 at $x = 4$ ◀ a square is never negative

STEP 4 $f(x) \geq 0$ ◀ (b) the range

ANSWER (a) 9 (b) $f(x) \geq 0$ ✓ every output is 0 or more ✓

WATCH OUT

✗ range: $x \geq 0$

✓ $f(x) \geq 0$

The range is about outputs, so write $f(x)$ (or y).

EXAM TIP

Excluded value of $\frac{11}{2x-5}$: $2x - 5 = 0$, $x = 2.5$
— a 1-mark gift.

Check yourself

calculator first · answers on page 17

1.1 Nov 2023 P2H

$g : x \mapsto \frac{5}{2x-9}$. Which value of x cannot be in any domain of g ?

Answer:

1.2 May 2023 P1H

$g(x) = \frac{11}{2x-5}$. What value of x must be excluded?

Answer:

1.3 Jun 2025 P1H

$f(x) = 9 - \sqrt{x}$, $x \geq 0$. Find $f(9)$.

Answer:

V2**VARIANT 2**

Composite Functions

$fg(x)$: put $g(x)$ into f — the right-hand letter acts first.

SEEN IN PAST PAPERS

8 / 39

PRACTICE QUESTIONS: 14



Your notes

Came up in: Jan 2020 P1HR Q17(c) • Oct/Nov 2020 P1HR Q16(b) • Jan 2021 P2HR Q17(a) • May 2021 P1H Q24(a) • Nov 2021 P2H Q15(b) • Nov 2023 P2H Q19(b) • Jun 2025 P2HR Q19(b) • Nov 2025 P2H Q19(b)

? What is it?

In plain words

$gf(x)$ means **f first, then g** . With a number: work out $f(\text{number})$, then put that into g .

With x : write $g(\dots)$ and put the **whole of $f(x)$** in place of every x in g . Then simplify.

IT LOOKS LIKE

$$fg(2) \quad gf(x) \text{ for } f = 2x - 3$$

$$g = \frac{x}{3x+1}$$

$$g(x) = \frac{x}{3x+1}$$

$$gf(x) = \frac{(2x-3)}{3(2x-3)+1}$$

$$= \frac{2x-3}{6x-8}$$

How do I do it?

1 Order

Right-hand letter first: $gf = f$ then g .

2 Inner

Number: work it out. Letter: write $f(x)$ in a bracket.

3 Outer

Replace every x of the outer function by that bracket.

4 Simplify

Expand and collect.

RIGHT-HAND LETTER FIRST

$$fg(x) = f(g(x)).$$

V2 Composite Functions

VARIANT 2 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“Find $fg(2)$ ” → $g(2)$ **first**, then f of that

“Find $gf(x)$. Give your answer in its simplest form” → **Bracket $f(x)$ into g ; simplify**

“ $h(x) = fg(x)$ ” → **Write h as one expression first**

**WORKED EXAMPLE**

PAST PAPER

November 2021 · Paper 2H · Q15(b) · 2 marks

$f(x) = 2x - 3$ and $g(x) = \frac{x}{3x + 1}$. Find $gf(x)$. Simplify your answer.

STEP 1 $gf(x) = g(2x - 3)$ ◀ f first

STEP 2 $= \frac{(2x - 3)}{3(2x - 3) + 1}$ ◀ every x in g becomes $(2x - 3)$

STEP 3 $= \frac{2x - 3}{6x - 9 + 1}$ ◀ expand

STEP 4 $= \frac{2x - 3}{6x - 8}$ ◀ simplify

ANSWER $gf(x) = \frac{2x - 3}{6x - 8}$

✓ $x = 2$: $f = 1$, $g(1) = \frac{1}{4}$; formula $\frac{1}{4}$ ✓

WATCH OUT

✗ $gf(x) = f(g(x))$

✓ $gf(x) = g(f(x))$

The letter next to x acts first.

EXAM TIP

Test your $gf(x)$ with $x = 2$ both ways — it takes 20 seconds.

Check yourself

calculator first · answers on page 17

2.1

Jan 2020 P1HR

$f(x) = (x - 4)^2$,
 $g(x) = \frac{4}{x + 3}$. Work out
 $fg(2)$.

Answer:

2.2

May 2021 P1H

$f(x) = 5x^2 - 10x + 7$,
 $g(x) = 7x - 6$. Find $fg(2)$.

Answer:

2.3

Nov 2025 P2H

$f(x) = 3x - 2$,
 $g(x) = \frac{x}{2x - 1}$. Find
 $gf(x)$ in its simplest
form.

Answer:

V3

VARIANT 3

Solve with Composite Functions

Build $fg(x)$ as one expression, set it equal (or $>$, $<$), then solve.

SEEN IN PAST PAPERS

6 / 39

PRACTICE QUESTIONS: 9



Your notes

Came up in: Oct/Nov 2020 P1HR Q16(c) • Jan 2021 P2HR Q17(b) • Jun 2022 P2HR Q19 • May 2023 P1H Q17(b) • Jun 2024 P2HR Q17 • Jun 2025 P1H Q16(b)

? What is it?

In plain words

First write the composite as **one expression** (V2). Then it is an ordinary equation — linear, fraction or quadratic — or an **inequality**.

A fraction equation:

multiply by the bottom first.

IT LOOKS LIKE

$$fg(x) = 2$$

$$fg(k) = 2$$

$$fg(x) > 0$$

$$fg(x) = 5\left(\frac{5x}{x+4}\right) - 7 = 2$$

$$\frac{25x}{x+4} = 9$$

$$25x = 9x + 36$$

$$x = \frac{9}{4}$$

How do I do it?

1 Build

 $fg(x)$ as one expression.

2 Set up

 $=$ the value (or $>$, $<$).

3 Clear

Multiply by any bottom; expand.

4 Solve

Linear, quadratic or inequality (Chapter 2).

BUILD, THEN SOLVEWrite $fg(x)$ before the $=$ sign.

V3 Solve with Composite Functions

VARIANT 3 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO“Solve $fg(x) = 2$ ”→ **One expression, = 2, solve**“Given that $fg(k) = 2$, work out k ”→ **Same, with k for x** “Solve $fg(x) > 0$. Show clear algebraic working”→ **Quadratic inequality: critical values, sketch****WORKED EXAMPLE PAST PAPER**

Oct/Nov 2020 • Paper 1HR • Q16(c) • 3 marks

 $f : x \mapsto 5x - 7$ and $g : x \mapsto \frac{5x}{x+4}$. Solve $fg(x) = 2$.**STEP 1**

$$fg(x) = 5\left(\frac{5x}{x+4}\right) - 7$$

◀ build: g first**STEP 2**

$$\frac{25x}{x+4} - 7 = 2 \Rightarrow \frac{25x}{x+4} = 9$$

◀ set up

STEP 3

$$25x = 9(x+4) = 9x + 36$$

◀ multiply by the bottom

STEP 4

$$16x = 36 \Rightarrow x = \frac{9}{4}$$

◀ solve

ANSWER $x = \frac{9}{4}$

$$\checkmark g\left(\frac{9}{4}\right) = \frac{9}{5}, f\left(\frac{9}{5}\right) = 9 - 7 = 2 \checkmark$$

**WATCH OUT**

$$\times 25x = 9x + 4$$

$$\checkmark 25x = 9(x+4)$$

Multiply the whole bottom by 9 — bracket it.

**EXAM TIP**Inequalities like $fg(x) > 0$: find where $= 0$, then sketch to pick the regions.**Check yourself**

calculator first • answers on page 17

3.1

Jan 2021 P2HR

$$f(x) = x^2 + 6,$$

$$g(x) = x - 10. \text{ Solve}$$

$$fg(x) = f(x).$$

Answer:

3.2

Jun 2024 P2HR

$$f(x) = \frac{x}{2x-4},$$

$$g(x) = 3x+1. \text{ Given}$$

$$fg(k) = 2, \text{ find } k.$$

Answer:

3.3

Jun 2022 P2HR

$$f(x) = x^2 - 4,$$

$$g(x) = 2x + 1. \text{ Solve}$$

$$fg(x) > 0.$$

Answer:

V4

VARIANT 4

Inverse: Linear & Fractions

Write $y =$, rearrange for x (collect x terms, factorise), then swap.

SEEN IN PAST PAPERS

5 / 39

PRACTICE QUESTIONS: 5

 Your notes

Came up in: Oct/Nov 2020 P1HR Q16(d) • Jan 2021 P2HR Q17(d) • Nov 2021 P2H Q15(c) • Jun 2022 P1H Q14(b) • May 2024 P1H Q15(b)

? What is it?

In plain words

The inverse **undoes** the function.

Write $y = f(x)$, make x **the subject**, then rename: $x \rightarrow f^{-1}(x)$, $y \rightarrow x$.

With a fraction: multiply out, get the **x terms on one side**, **factorise x out** — exactly as in Changing the Subject (Chapter 2).

$$y(x - 2) = 3x + 1$$

$$xy - 3x = 2y + 1$$

$$x(y - 3) = 2y + 1$$

$$f^{-1}(x) = \frac{2x + 1}{x - 3}$$

IT LOOKS LIKE

$$\frac{3x+1}{x-2}$$

$$\frac{2x}{x-6}$$

$$5x - 7$$

How do I do it?

1 $y =$

$$y = \frac{3x+1}{x-2}$$

2 Clear

Multiply by the bottom; expand.

3 x subject x terms one side, factorise x , divide.**4** SwapWrite in terms of x : $f^{-1}(x) = \dots$

$y =$, REARRANGE, SWAP Factorise x out when it appears twice.

V4 Inverse: Linear & Fractions

VARIANT 4 · SEE IT DONE, THEN TRY


 Your notes
WHEN YOU SEE → YOU DO

“Express the inverse function f^{-1} in the form $f^{-1}(x) = \dots$ ”

→ $y =$, make x the subject, swap

x on the top and the bottom

→ Multiply out, collect, factorise x

“in the form $f^{-1} : x \mapsto \dots$ ”

→ Same answer, written with $\mapsto$ **WORKED EXAMPLE** **PAST PAPER**

May 2024 · Paper 1H · Q15(b) · 3 marks

$f : x \mapsto \frac{3x+1}{x-2}$. Express the inverse function f^{-1} in the form $f^{-1}(x) = \dots$

STEP 1 $y = \frac{3x+1}{x-2} \Rightarrow y(x-2) = 3x+1$ ◀ $y =$, clear the bottom

STEP 2 $xy - 2y = 3x + 1 \Rightarrow xy - 3x = 2y + 1$ ◀ collect x terms

STEP 3 $x(y-3) = 2y+1 \Rightarrow x = \frac{2y+1}{y-3}$ ◀ factorise, divide

STEP 4 $f^{-1}(x) = \frac{2x+1}{x-3}$ ◀ swap

ANSWER $f^{-1}(x) = \frac{2x+1}{x-3}$

✓ $f(3) = 10$, $f^{-1}(10) = \frac{21}{7} = 3$ ✓

WATCH OUT

✗ $f^{-1}(x) = \frac{x-2}{3x+1}$

✓ rearrange for x

The inverse is not the fraction turned upside down.

EXAM TIP

Test with one number: $f(a) = b$ must give $f^{-1}(b) = a$.

Check yourself

calculator first · answers on page 17

4.1 Jun 2022 P1H

$f : x \mapsto \frac{2x}{x-6}$. Express
 $f^{-1} : x \mapsto \dots$

Answer:

4.2 Oct/Nov 2020 P1HR

$g : x \mapsto \frac{5x}{x+4}$. Express
 $g^{-1} : x \mapsto \dots$

Answer:

4.3 Jan 2021 P2HR

$h(x) = \frac{2x-4}{x}$. Express
 $h^{-1}(x) = \dots$

Answer:

V5

VARIANT 5

Inverse of a Quadratic

Complete the square, rearrange, and let the domain choose the + or – root.

SEEN IN PAST PAPERS

7 / 39

PRACTICE QUESTIONS: 2



Your notes

Came up in: Jan 2020 P1H Q21 • Jan 2021 P1H Q22 • May 2021 P1H Q24(b) • Jan 2022 P2HR Q23 • Jan 2022 P2H Q25(a) • Nov 2023 P2H Q19(c) • Nov 2024 P1H Q25

? What is it?**In plain words**x appears twice (x^2 and x), so first**complete the square:** $3x^2 - 12x + 8 = 3(x - 2)^2 - 4$. Now x appears once.Rearrange; at the square root choose the sign from the **domain**: $x > 2$ means $x = 2 + \sqrt{\dots}$ **IT LOOKS LIKE**

$x^2 - 8x + 5$

$x \leq 4; 3x^2 - 12x + 8$

$x > 2$

$$y = 3(x - 2)^2 - 4$$

$$(x - 2)^2 = \frac{y + 4}{3}$$

$$x - 2 = \pm \sqrt{\frac{y + 4}{3}} \quad (x > 2: +)$$

$$h^{-1}(x) = 2 + \sqrt{\frac{x + 4}{3}}$$

How do I do it?**1 Complete the square**

$$a(x - p)^2 + q.$$

2 Rearrange

$$(x - p)^2 = \frac{y - q}{a}.$$

3 Root, choose sign

$$x \geq p: +. \quad x \leq p: -.$$

4 Swap

$$\text{Write } f^{-1}(x) = p \pm \sqrt{\dots}$$

SQUARE, ROOT, DOMAIN PICKS THE SIGN 7 of 39 parts.

V5 Inverse of a Quadratic

VARIANT 5 • SEE IT DONE, THEN TRY

 Your notes
WHEN YOU SEE → YOU DO

" $f(x) = x^2 - 8x + 5$ where $x \leq 4$. Express f^{-1} "

→ **Complete the square; $x \leq 4 \rightarrow -$ root**

"State the domain of g^{-1} "

→ **Domain of g^{-1} = range of g** **WORKED EXAMPLE****PAST PAPER**

November 2023 • Paper 2H • Q19(c) • 3 marks

$h : x \mapsto 3x^2 - 12x + 8$ where $x > 2$. Express the inverse function h^{-1} in the form $h^{-1} : x \mapsto \dots$

STEP 1 $3x^2 - 12x + 8 = 3(x - 2)^2 - 4$ ◀ complete the square

STEP 2 $y = 3(x - 2)^2 - 4 \Rightarrow (x - 2)^2 = \frac{y + 4}{3}$ ◀ rearrange

STEP 3+4 $x = 2 + \sqrt{\frac{y+4}{3}} \Rightarrow h^{-1} : x \mapsto 2 + \sqrt{\frac{x+4}{3}}$ ◀ $x > 2$: + root; swap

ANSWER $h^{-1} : x \mapsto 2 + \sqrt{\frac{x+4}{3}}$

✓ $h(3) = -1$; $2 + \sqrt{1} = 3$ ✓

WATCH OUT

$$\times 2 \pm \sqrt{\frac{x+4}{3}}$$

$$\checkmark 2 + \sqrt{\frac{x+4}{3}}$$

An inverse function gives one answer: the domain picks the sign.

EXAM TIP

Check the completed square by expanding it back before you go on.

Check yourself

calculator first • answers on page 17

5.1

Jan 2021 P1H

$f(x) = x^2 - 8x + 5$ where $x \leq 4$. Express $f^{-1}(x) = \dots$

Answer:

5.2

Jan 2020 P1H

$f(x) = x^2 - 2x$,
 $g(x) = x + 3$, $h(x) = fg(x)$
for $x \geq -2$. Express $h^{-1}(x)$.

Answer:

5.3

Nov 2024 P1H

$f(x) = 2x^2 - 24x + 7$
where $x \geq 6$. Find $f^{-1}(x)$.

Answer:

V6

VARIANT 6

Function Machines

Each box is one operation; the inverse runs the boxes backwards with opposite operations.

SEEN IN PAST PAPERS

0 / 39

NOT ASKED YET · BE READY
PRACTICE QUESTIONS: 7

 Your notes

? What is it?

In plain words

A function machine shows the **order of operations**: $\times 5$ then -10 is $f(x) = 5x - 10$.

To go back (inverse, or “find the input”), **reverse the order** and use the **opposite** of each box.

IT LOOKS LIKE

 $\times 5 \rightarrow -10$ square $\rightarrow +6$

$$x \rightarrow \boxed{\times 5} \rightarrow \boxed{-10} \rightarrow 5x - 10$$

$$\text{reverse: } x \rightarrow \boxed{+10} \rightarrow \boxed{\div 5} \rightarrow \frac{x + 10}{5}$$

How do I do it?

1 Read

Boxes left to right, in order.

2 Write

Build the expression with brackets where needed.

3 Reverse

Opposite operations, opposite order.

4 Square?Undo with $\pm\sqrt{}$ — two inputs possible.**OPPOSITE OPS, OPPOSITE ORDER**

Like taking off socks and shoes.

V6 Function Machines

VARIANT 6 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

A diagram: input $\rightarrow [\times 3] \rightarrow [+4] \rightarrow$ output $\rightarrow f(x) = 3x + 4$

"Find the input when the output is 19" $\rightarrow$ **Reverse machine:** $-4, \div 3$

"Use the graph to find $f(-1)$ " $\rightarrow$ **Read y at $x = -1$**

WORKED EXAMPLE **DR ESLAM**

Function machine · 3 marks

A machine does $\times 5$ then -10 . (a) Write the function $f(x)$. (b) Find $f^{-1}(x)$. (c) Find the input when the output is 15.

STEP 1 $f(x) = 5x - 10$

◀ (a) boxes in order

STEP 2 $+10$ then $\div 5$

◀ reverse the machine

STEP 3 $f^{-1}(x) = \frac{x + 10}{5}$

◀ (b)

STEP 4 $f^{-1}(15) = \frac{25}{5} = 5$

◀ (c)

ANSWER (a) $5x - 10$ (b) $\frac{x + 10}{5}$ (c) 5✓ $5 \times 5 - 10 = 15$ ✓**WATCH OUT**✗ reverse: $\div 5$ then $+10$ ✓ $+10$ then $\div 5$

Undo the last box first.

EXAM TIP

Write each reversed box above an arrow — order mistakes vanish.

Check yourself

calculator first · answers on page 17

6.1 Dr Eslam

A machine: $\times 3$ then $+4$. Write $f(x)$ and find the input for output 19.

Answer:

6.2 Dr Eslam

$f(x) = 2(x - 3)$. Use a reverse machine to find $f^{-1}(x)$.

Answer:

6.3 Dr Eslam

A machine: square, then $+6$. The output is 15. Find the inputs.

Answer:

CHECK YOURSELF

Answers

Every answer below was checked with sympy (check_answers.py).

$$1.1 \quad x = 4.5$$

$$1.2 \quad x = 2.5$$

$$1.3 \quad 6$$

$$2.1 \quad \frac{256}{25}$$

$$2.2 \quad 247$$

$$2.3 \quad \frac{3x - 2}{6x - 5}$$

$$3.1 \quad x = 5$$

$$3.2 \quad k = \frac{5}{9}$$

$$3.3 \quad x < -1.5 \text{ or } x > 0.5$$

$$4.1 \quad \frac{6x}{x - 2}$$

$$4.2 \quad \frac{4x}{5 - x}$$

$$4.3 \quad \frac{4}{2 - x}$$

$$5.1 \quad 4 - \sqrt{x + 11}$$

$$5.2 \quad -2 + \sqrt{x + 1}$$

$$5.3 \quad 6 + \sqrt{\frac{x + 65}{2}}$$

$$6.1 \quad 3x + 4; \text{ input } 5$$

$$6.2 \quad \frac{x}{2} + 3$$

$$6.3 \quad 3 \text{ or } -3$$

THE WHOLE TOPIC ON ONE PAGE

Functions cheat sheet



Your notes
V1 Domain, range 13/39

$$2x - 9 \neq 0$$

Bottom $\neq 0$; squares ≥ 0 .**V2 Composite 8/39**

$$gf(x) = g(f(x))$$

Right-hand letter first.

V3 Solve with fg 6/39

$$\frac{25x}{x+4} = 9$$

Build, then solve.

V4 Inverse: fraction 5/39

$$x(y - 3) = 2y + 1$$

 $y =$, rearrange, swap.**V5 Inverse: quadratic 7/39**

$$2 + \sqrt{\frac{x+4}{3}}$$

Complete the square; domain picks the sign.

V6 Machines+10, $\div 5$

Opposite ops, opposite order.

**TOP 3 LOST MARKS**

1. fg done in the wrong order $\times$ 2. $\pm$ left in an inverse $\times$ 3. range written with $x \times$

BOTTOM LINE Substitute every x · bottom $\neq 0$ · right-hand letter first · $y =$, rearrange, swap.